EXPERIMENT :-7

CODE:-

```
#include <stdio.h>

int a[20], x[20];
int n, d;

void subset(int k, int sum) {
  int i;

  if (sum == d) {
    printf("Subset: ");
    for (i = 0; i < k; i++)
      if (x[i])
        printf("%d ", a[i]);
    printf("\n");
    return;
  }

  if (k == n || sum > d)
    return;

  // Include the element
  x[k] = 1;
  subset(k + 1, sum + a[k]);

  // Exclude the element
```

```
    x[k] = 0;
    subset(k + 1, sum);
}

int main() {
    int i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter required sum: ");
    scanf("%d", &d);

    printf("\nSubsets are:\n");
    subset(0, 0);

    return 0;
}
```

Example Input

```
Enter number of elements: 4
Enter elements:
2 4 6 8
Enter required sum: 10
```

Output

```
Subsets are:
Subset: 2 8
Subset: 4 6
```